

21/04/2025



Aakash

Medical | IIT-JEE | Foundations

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CODE-A



AIM - 720

(Advanced **INTENSIVE** Mastery for 720)

MM : 720

CST-8

Time : 180 Mins.

Answers

1. (4)	37. (1)	73. (3)	109. (4)	145. (2)
2. (1)	38. (2)	74. (3)	110. (3)	146. (3)
3. (3)	39. (4)	75. (3)	111. (2)	147. (1)
4. (2)	40. (2)	76. (4)	112. (2)	148. (2)
5. (4)	41. (2)	77. (1)	113. (4)	149. (4)
6. (4)	42. (2)	78. (4)	114. (1)	150. (3)
7. (1)	43. (1)	79. (2)	115. (3)	151. (3)
8. (3)	44. (2)	80. (4)	116. (2)	152. (4)
9. (3)	45. (3)	81. (3)	117. (4)	153. (3)
10. (1)	46. (4)	82. (1)	118. (2)	154. (3)
11. (1)	47. (1)	83. (2)	119. (4)	155. (2)
12. (2)	48. (2)	84. (3)	120. (4)	156. (2)
13. (1)	49. (4)	85. (3)	121. (3)	157. (4)
14. (4)	50. (3)	86. (1)	122. (3)	158. (4)
15. (3)	51. (2)	87. (3)	123. (2)	159. (3)
16. (3)	52. (3)	88. (4)	124. (4)	160. (4)
17. (3)	53. (1)	89. (4)	125. (1)	161. (2)
18. (2)	54. (2)	90. (1)	126. (2)	162. (1)
19. (3)	55. (1)	91. (2)	127. (1)	163. (2)
20. (1)	56. (2)	92. (3)	128. (3)	164. (3)
21. (3)	57. (2)	93. (4)	129. (2)	165. (1)
22. (3)	58. (2)	94. (1)	130. (3)	166. (2)
23. (2)	59. (1)	95. (4)	131. (1)	167. (3)
24. (3)	60. (1)	96. (3)	132. (2)	168. (2)
25. (4)	61. (1)	97. (2)	133. (1)	169. (2)
26. (1)	62. (4)	98. (3)	134. (2)	170. (4)
27. (3)	63. (4)	99. (1)	135. (1)	171. (4)
28. (1)	64. (3)	100. (1)	136. (2)	172. (3)
29. (1)	65. (3)	101. (2)	137. (4)	173. (3)
30. (1)	66. (3)	102. (1)	138. (3)	174. (2)
31. (3)	67. (1)	103. (1)	139. (1)	175. (1)
32. (4)	68. (4)	104. (3)	140. (4)	176. (2)
33. (3)	69. (2)	105. (1)	141. (4)	177. (4)
34. (4)	70. (3)	106. (4)	142. (3)	178. (3)
35. (3)	71. (4)	107. (4)	143. (2)	179. (1)
36. (2)	72. (4)	108. (4)	144. (4)	180. (2)

Hints and Solutions

CHEMISTRY

(1) Answer : (4)

Solution:

 Iron catalysts are used in Haber's process. In wacker's process, oxidation of ethene to ethanol is catalysed by PdCl_2 .

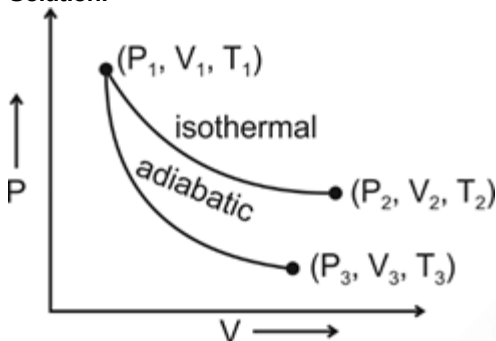
(2) Answer : (1)

Solution:

Interstitial compounds have high melting points, even higher than those of pure metals and are chemically inert.

(3) Answer : (3)

Solution:


 $\therefore W_{\text{isothermal}} > W_{\text{adiabatic}}$

(4) Answer : (2)

Solution:

1 molal solution will have 1 mol solute in 1000 g solvent

$$X_{\text{solute}} = \frac{1}{55.5+1} = \frac{1}{56.5} = 0.017$$

(5) Answer : (4)

Solution:

$$q = mC\Delta T$$

$$1000 \text{ J} = \frac{100}{18} \times 75 \times \Delta T$$

$$\frac{10 \times 18}{75} = \Delta T$$

$$2.4 \text{ K} = \Delta T$$

(6) Answer : (4)

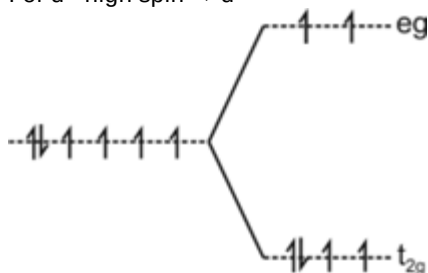
Hint:

For octahedral complexes

$$\text{CFSE} = -0.4 \Delta_0 \times n_{e t_{2g}} + 0.6 \Delta_0 \times n_{e e_g}$$

 $(n_e \Rightarrow \text{number of electrons})$

Solution:

 For d^6 high spin $\rightarrow d$


$$\text{CFSE} = -0.4 \Delta_0 \times 4 + 0.6 \Delta_0 \times 2 = -0.4 \Delta_0$$

(7) Answer : (1)

Solution:

Oxidation state of cobalt

$$+1 + x + 4 \times 0 = 0$$

$$x = -1$$

(8) Answer : (3)

Solution:

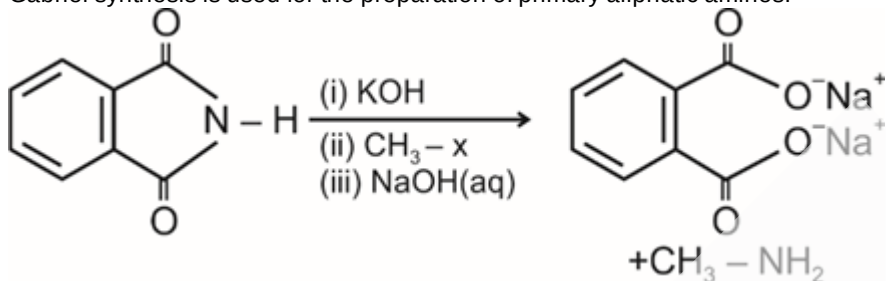
The basicity of amines in aqueous solution depend on the inductive effect, solution effect and steric hindrance of alkyl group

Correct order of basicity : $(\text{CH}_3)_2\text{NH} > \text{CH}_3\text{NH}_2 > \text{NH}_3 > (\text{CH}_3)_3\text{N}$

(9) Answer : (3)

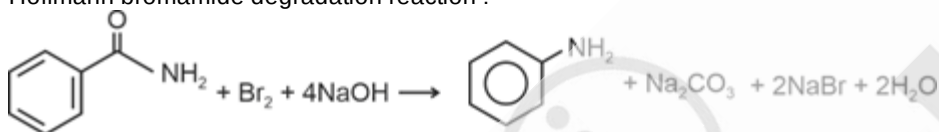
Solution:

Gabriel synthesis is used for the preparation of primary aliphatic amines.

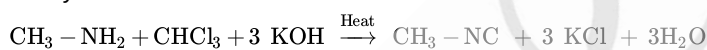


Above reaction is known as Gabriel phthalimide synthesis

Hoffmann bromamide degradation reaction :



Carbylamine reaction :



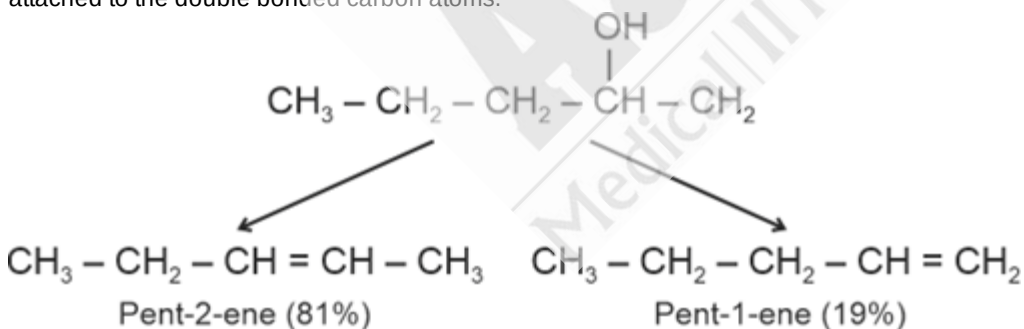
Sandmeyer reaction :



(10) Answer : (1)

Solution:

In dehydrohalogenation reactions the preferred product is that alkene which has the greater number of alkyl groups attached to the double bonded carbon atoms.

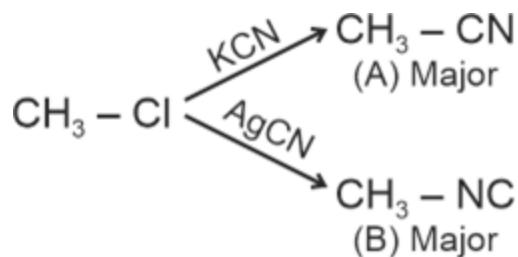


Since β -hydrogen atom is involved in elimination so it is often called β -elimination.

(11) Answer : (1)

Solution:

KCN is predominantly ionic and provides cyanide ions in solution AgCN is mainly covalent in nature.



(12) Answer : (2)

Solution:

Ionisation enthalpy : $\text{Na} \rightarrow \text{Na}^+ + e^-$;

$E = 5.1 \text{ eV}$

Electron gain enthalpy : $\text{Na}^+ + e^- \rightarrow \text{Na}$;

$E = -5.1 \text{ eV}$

(13) Answer : (1)

Solution:

A mixture of NH_4Cl and NH_4OH act as buffer solution.

(14) Answer : (4)

Solution:

In mercury cell;

Anode : Zn-Hg amalgam

Cathode : Paste of HgO and carbon

Electrolytes : Paste of KOH and ZnO .

(15) Answer : (3)

Solution:

Electronic configuration (atomic No = 60)

$= [\text{Xe}] 4f^4 6s^2$

(16) Answer : (3)

Solution:

$$\text{pH} = \text{pKa} + \log \frac{[\text{Salt}]}{[\text{Acid}]}$$

$$\text{pH} = 4.76 + \log \frac{(0.2)}{(0.1)}$$

$$\text{pH} = 4.76 + \log 2$$

$$\text{pH} = 4.76 + 0.3010$$

$$\text{pH} = 5.0610$$

$$\text{pOH} = 14 - 5.0610$$

$$\text{pOH} = 8.9390$$

(17) Answer : (3)

Solution:

Ions	Limiting molar conductivity (in $\text{S cm}^2 \text{ molar}^{-1}$)
Na^+	50.1
K^+	73.1
Ca^{2+}	119.0
Mg^{2+}	106.0

(18) Answer : (2)

Solution:

Fluorine does not show variable oxidation state.

(19) Answer : (3)

Solution:

The acidic character of group 16 hydrides increases from H_2O to H_2Te .

(20) Answer : (1)

Solution:

Hypobromous acid $\rightarrow \text{HOBr}^{+1}$

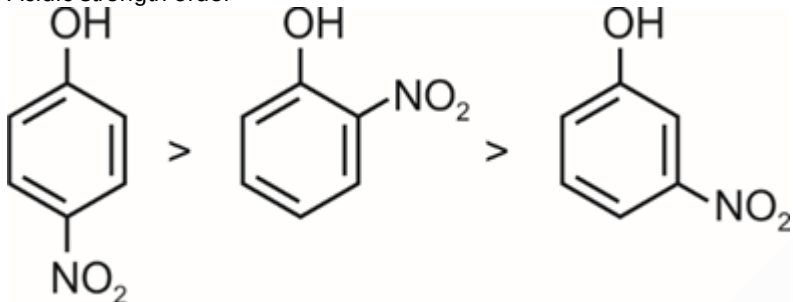
Perbromic acid $\rightarrow \text{HOBrO}_3^{+7}$

Sum of oxidation state of Br = $1 + 7 = 8$

(21) Answer : (3)

Solution:

Acidic strength order



Propanol (1° alcohol) will not give immediate turbidity with Lucas reagent.

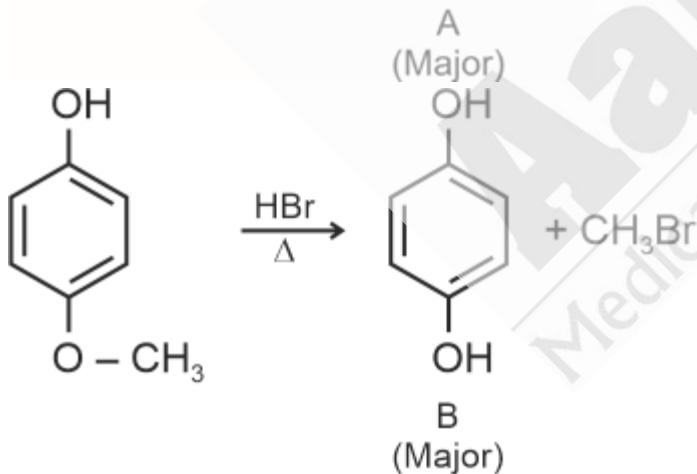
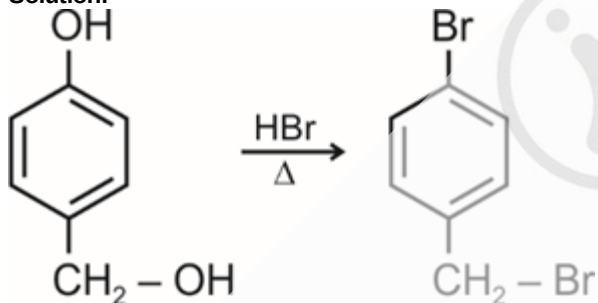
(22) Answer : (3)

Solution:

Due to Hydrogen bonding alcohol have highest boiling point.

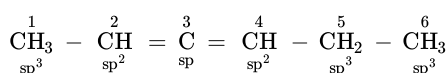
(23) Answer : (2)

Solution:



(24) Answer : (3)

Solution:



(25) Answer : (4)

Solution:

1 M of 10 mL $\text{H}_2\text{SO}_4 = 20 \text{ m mol } \text{H}_2\text{SO}_4$

mole of $\text{NH}_3 = \text{mole of } \text{H}_2\text{SO}_4$

= 20 m mol

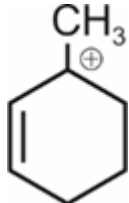
∴ 1000 mL of 1 M ammonia ≡ 14g NH₃

So, 20 mL of 1 M ammonia = $\frac{14 \times 20}{1000}$ g

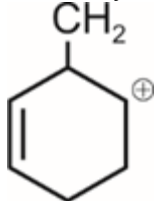
% of Nitrogen = $\frac{14 \times 20 \times 100}{1000 \times 0.56}$ = 50%

(26) Answer : (1)

Solution:



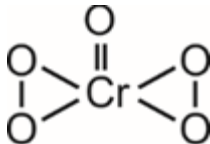
is a tertiary carbocation with more number of hyperconjugation structures (7α hydrogen atoms) while



is secondary carbocation with 3α hydrogen atoms and 3 hyperconjugative structure.

(27) Answer : (3)

Solution:



4 Oxygen atoms are involved in peroxy linkage

(28) Answer : (1)

Solution:

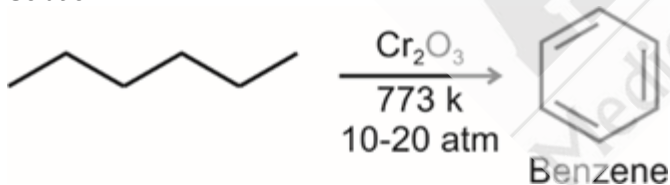


Sodium tetrahydroxo

-aluminate (III)

(29) Answer : (1)

Solution:



(30) Answer : (1)

Solution:

• Vitamin A, Vitamin D, Vitamin E, Vitamin K are fat soluble Vitamin

• Vitamin B₁ (Thiamine), Vitamin B₂ (Riboflavin), Vitamin B₆ (pyridoxine), Vitamin B₁₂ and Vitamin C are water soluble Vitamin

(31) Answer : (3)

Solution:

Orbital angular momentum = $\sqrt{l(l+1)} \hbar$

= $\sqrt{2 \times 3} \hbar = \sqrt{6} \hbar$

(32) Answer : (4)

Solution:



Compounds	Melting point (K)
2,2-Dimethylpropane	256.4
Butane	134.6
2-Methylpropane	114.7
2-methylbutane	113.1

(33) Answer : (3)

Solution:

Thyroxine produced in the thyroid gland is an iodinated derivative of amino acid tyrosin.

(34) Answer : (4)

Solution:

$$E = -2.18 \times 10^{-18} \frac{z^2}{n^2} \text{ J}$$

$$E = -2.18 \times 10^{-18} \times \frac{(3)^2}{(2)^2} \text{ J}$$

$$E = -4.9 \times 10^{-18} \text{ J}$$

(35) Answer : (3)

Solution:

$$\Rightarrow \sigma 1s^2 \sigma^* 1s^2 \sigma 2s^2 \sigma^* 2s^2 (\pi 2P_x^2 = \pi 2P_y^2) \sigma 2P_z^2$$

(36) Answer : (2)

Solution:

SO₃ has two π - π bonds.

(37) Answer : (1)

Solution:

In PCl₅ axial bonds are longer than equatorial bonds.

(38) Answer : (2)

Solution:

$$P_{\text{total}} = P_1 + P_2$$

$$= X_1 P_1^0 + x_2 P_2^0$$

$$= \frac{1}{4} \times 360 + \frac{3}{4} \times 300$$

$$= 90 + 225 = 315 \text{ mmHg}$$

Mole fraction of hexane in vapour phase.

$$\frac{P}{P_{\text{total}}} = \frac{x_1 P_1^0}{P_T} = \frac{90}{315}$$

(39) Answer : (4)

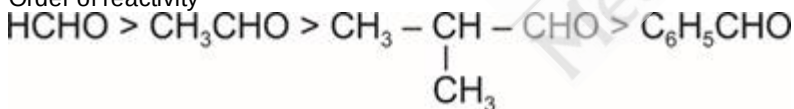
Solution:

Smaller is the value of Henry's constant higher will be the solubility of gas.

(40) Answer : (2)

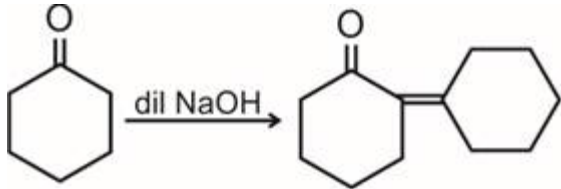
Solution:

Order of reactivity



(41) Answer : (2)

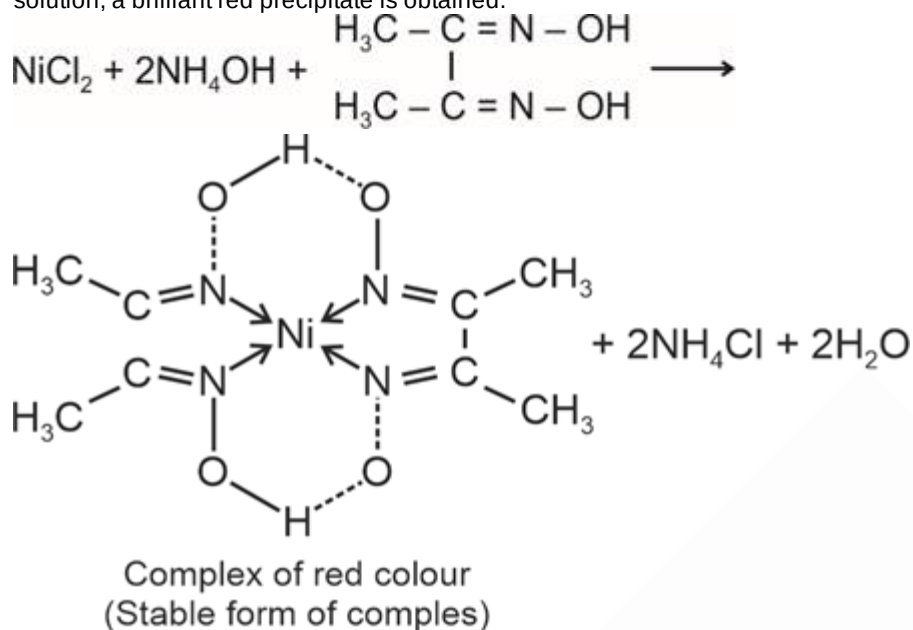
Solution:



(42) Answer : (2)

Solution:

When dimethyl glyoxime is added to the aqueous solution of nickel chloride and is made alkaline by adding NH_4OH solution, a brilliant red precipitate is obtained.



(43) Answer : (1)

Solution:

For zero order reaction $t_{100\%} = \frac{a}{k}$

(44) Answer : (2)

Solution:

$\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$ – Prussian blue

PbS – Black

$\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$ – Green

$\text{Fe}(\text{SCN})_3$ – Blood red

(45) Answer : (3)

Solution:

Unit of rate of reaction is $\text{mol l}^{-1} \text{s}^{-1}$ irrespective of order of reaction.

PHYSICS

(46) Answer : (4)

Solution:

- In uniform circular motion speed of object remains constant while acceleration changes continuously.
- Speed is a scalar quantity while velocity and acceleration are both vector quantities.

(47) Answer : (1)

Solution:



In isochoric P - V diagram, volume on graph-line remains same.

(48) Answer : (2)

Solution:

No. of possible spectral lines emitted = $\frac{n(n-1)}{2}$

$$= \frac{5(4)}{2} = 10$$

(49) Answer : (4)

Solution:

13th group elements act as acceptor impurity.

(50) Answer : (3)

Solution:

$$Y = \overline{\overline{A} + BC}$$

$$= \overline{\overline{A} \cdot \overline{BC}}$$

$$A \cdot (\overline{B} + \overline{C})$$

(51) Answer : (2)

Solution:

$${}_1p^1 \rightarrow {}_0n^1 + e^+ + \nu$$

Therefore, $\frac{N}{P}$ ratio increases.

(52) Answer : (3)

Solution:

$$R = \frac{V}{I}$$

$$= \frac{1}{2} = 0.5 \Omega$$

(53) Answer : (1)

Solution:

Heat is a form of energy, so its dimensional formula is same as energy i.e., $[ML^2T^{-2}]$.

(54) Answer : (2)

Solution:

$$[L] = [bT^2]$$

$$[b] = [LT^{-2}]$$

(55) Answer : (1)

Solution:

Acceleration = Slope of v - t curve

For A, slope = $\tan 50^\circ$

For B, slope = $\tan 20^\circ$

$$\text{Ratio} = \frac{\tan 50^\circ}{\tan 20^\circ}$$

(56) Answer : (2)

Solution:

$$S_4 = 9 - \frac{2}{2} (2 \times 4 - 1)$$

$$= 9 - 7 = 2 \text{ m}$$

(57) Answer : (2)

Solution:

$$1 \times 3\hat{i} + 1 \times 4\hat{j} + 1 \times \vec{v}_3 = \text{zero}$$

$$\vec{v}_3 = -(3\hat{i} + 4\hat{j}) \text{ m/s}$$

$$F_{\text{avg}} = \frac{-(3\hat{i} + 4\hat{j})}{10^{-4}} = -(3\hat{i} + 4\hat{j}) \times 10^4 \text{ N}$$

(58) Answer : (2)

Solution:

$$\theta = \tan^{-1} \left(\frac{v^2}{rg} \right)$$

$$= \tan^{-1} \left(\frac{400}{40\sqrt{3} \times 10} \right)$$

$$\theta = 30^\circ$$

(59) Answer : (1)

Solution:

$$\vec{A} \cdot \vec{B} = 0$$

$$2 + 2c = 0$$

$$2c = -2$$

$$c = -1$$

(60) Answer : (1)

Solution:

Work done along a closed path is zero for conservative forces.

(61) Answer : (1)

Solution:

Mutual inductance and self-inductance have same unit.

(62) Answer : (4)

Solution:



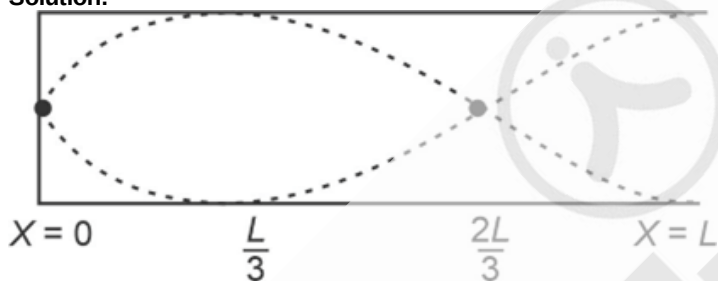
$$v_{\max} = 12 \text{ cm/s} = A\omega$$

$$\omega = \frac{12}{2} = 6 \text{ rad/s} = 2\pi f$$

$$\frac{6}{2\pi} = f = \frac{3}{\pi} \text{ Hz}$$

(63) Answer : (4)

Solution:



$$\text{Displacement antinode } X = \frac{L}{3}$$

$$\text{Displacement node } X = \frac{2L}{3}$$

$$\frac{3\lambda}{4} = L$$

$$\lambda = \frac{4L}{3}$$

(64) Answer : (3)

Solution:

They (EM waves) travel with different speed in different medium.

(65) Answer : (3)

Solution:

$$\text{Ratio of amplitudes} = \frac{a}{b} = \frac{3}{5}$$

$$\frac{I_{\max}}{I_{\min}} = \frac{(a+b)^2}{(a-b)^2}$$

$$= \frac{(8)^2}{(2)^2} = \frac{16}{1}$$

(66) Answer : (3)

Solution:

$$M = NIA$$

$$= I \frac{3}{4} \pi R^2$$

$$= \frac{3}{4} \pi R^2 I$$

(67) Answer : (1)

Solution:

For voltmeter, a high resistance is connected in series with galvanometer.

For ammeter, a low resistance is connected in parallel with galvanometer.

(68) Answer : (4)

Solution:

$$W = U_f - U_i$$

$$U_i = -MB$$

$$U_f = -MB \cos 53^\circ$$

$$= \frac{-3MB}{5}$$

$$W = \frac{-3MB}{5} + MB$$

$$4 \text{ J} = \frac{2MB}{5} \Rightarrow MB = 10 \text{ N m}$$

$$\tau = MB \sin \theta$$

$$= MB \sin 53^\circ$$

$$= \frac{4}{5} \times 10 = 8 \text{ N m}$$

(69) Answer : (2)

Solution:

$$X_L = \omega L$$

$$= 500 \times 40 \times 10^{-3} = 20 \Omega$$

$$I = \frac{V}{X_L} = \frac{40}{20} = 2 \text{ A}$$

(70) Answer : (3)

Solution:

$$\varepsilon = Blv_{\perp}$$

$$= 4 \times 2 \times 0.6 \times \cos 60^\circ = 2.4 \text{ V}$$

(71) Answer : (4)

Solution:

Transformer works for A.C. source and not for D.C. source.

(72) Answer : (4)

Solution:

Stress vs strain graph for an elastomer is not a straight line (rather close to parabola).

Steel is not an elastomer.

(73) Answer : (3)

Solution:

Attractive force acting on both of them = F

$$F = \frac{G(3m)(5m)}{\left(\frac{d}{4}\right)^2} = \frac{15Gm^2}{d^2} \times 16$$

$$a_{3m} = \frac{F}{3m} = \frac{80Gm^2}{d^2}$$

$$a_{5m} = \frac{F}{5m} = \frac{48Gm^2}{d^2}$$

a_{com} is zero as the gravitational force is an internal force in the system.

(74) Answer : (3)

Solution:

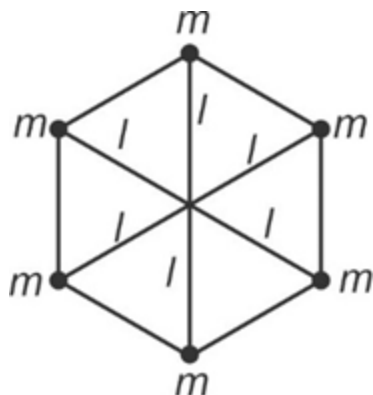
$$\text{Angular momentum } \vec{L} = \vec{r} \times \vec{p} = \text{constant}$$

$$\therefore \text{Torque } \vec{\tau} = \text{zero and } \vec{\alpha} = \text{zero}$$

(75) Answer : (3)

Solution:

Each of the six triangles in figure is an equilateral triangle.



$$V_{\text{centre}} = \left(\frac{-Gm}{l} \right) \times 6 = \frac{-6Gm}{l}$$

(76) Answer : (4)

Solution:

For earth revolving around sun,

$$\vec{L} = \vec{r} \times m\vec{v} = \text{constant}$$

$$\therefore |\vec{r} \times \vec{v}| = \text{same at A, B, C.}$$

(77) Answer : (1)

Solution:

Modern reflecting telescopes use a concave mirror for objective because it is rather difficult and expensive to make large sized lenses which form images free from chromatic aberration.

(78) Answer : (4)

Solution:

When the bird is seen by fish (in denser medium)

$$\Rightarrow h_1 = \mu_{\text{water}} (\text{real height}) = \frac{4}{3} h_2$$

$$\therefore \frac{h_1}{h_2} = \frac{4}{3}$$

(79) Answer : (2)

Solution:

Resistivity is the property of material, while resistance of any conductor depends on the dimensions too. Resistivity is a function of temperature.

(80) Answer : (4)

Solution:

$$C_{eq} = \frac{A\epsilon_0}{\frac{t_1}{k_1} + \frac{t_2}{k_2}}$$

$$C_{eq} = \frac{A\epsilon_0}{\frac{t}{K} + \frac{(d-t)}{1}} = \frac{A\epsilon_0 K}{Kd + t(1-K)}$$

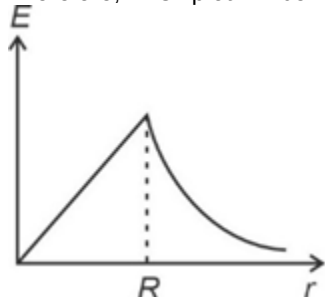
(81) Answer : (3)

Solution:

$$\text{For } r < R, E = \frac{kQr}{R^3}$$

$$\text{For } r \geq R, E = \frac{kQ}{r^2}$$

Therefore, E vs r plot will be



(82) Answer : (1)

Solution:

$$\vec{E} = -\frac{\partial V}{\partial x}\hat{i} - \frac{\partial V}{\partial y}\hat{j} - \frac{\partial V}{\partial z}\hat{k}$$

$$\vec{E} = -\left(-\frac{1}{x^2}\right)\hat{i} - \left(-\frac{1}{y^2}\right)\hat{j}$$

At point (1, 1):

$$\vec{E} = \frac{1}{1}\hat{i} + \frac{1}{1}\hat{j}$$

$$\vec{E} = (\hat{i} + \hat{j}) \text{ V/m}$$

(83) Answer : (2)

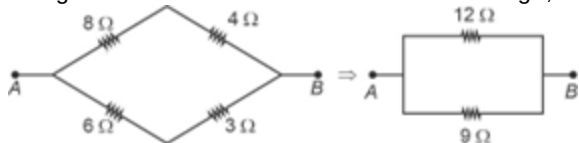
Solution:

Electrostatic force between two charges is independent of presence of any other charge.

(84) Answer : (3)

Solution:

The given circuit is a balanced wheatstone bridge, hence the simplified circuit will be



$$R_{eq} = \frac{R_1 R_2}{R_1 + R_2} = \frac{12 \times 9}{21} = \frac{36}{7} \Omega$$

(85) Answer : (3)

Solution:

Bernoulli's equation ideally applies to fluids with zero viscosity.

Steady flow is achieved at low flow speeds while beyond a limiting value this flow loses steadiness and becomes turbulent.

(86) Answer : (1)

Solution:

$$\tan \theta = \frac{a}{g} = \frac{g}{4g} = \frac{1}{4}$$

$$\Rightarrow \theta = \tan^{-1} \left(\frac{1}{4} \right)$$

(87) Answer : (3)

Solution:

Rate of heat loss is given by

$$\frac{dQ}{dt} = e\sigma A(T^4 - T_0^4)$$

$$\Rightarrow \frac{dQ}{dt} \propto A \propto r^2$$

(88) Answer : (4)

Solution:

$$PV^\gamma = \text{constant}$$

$$\Rightarrow P_1 V_1^\gamma = P_2 V_2^\gamma$$

$$\Rightarrow \frac{P_2}{P_1} = \left(\frac{V_1}{V_2} \right)^\gamma$$

$$\Rightarrow \frac{P_2}{P} = \left(\frac{27}{1} \right)^{\frac{5}{3}} = 3^5 = 243P$$

(89) Answer : (4)

Solution:

• If absorbed energy of photon is more than a threshold energy, then electron will emit otherwise not. Hence generally number of emitted electrons is less than number of incident photons

• Photoelectric effect can be explained by particle nature of light.

(90) Answer : (1)

Solution:

$$\text{Mean free path } \lambda = \frac{1}{\sqrt{2} n \pi d^2} \Rightarrow \lambda \propto \frac{1}{n}$$

BIOLOGY

(91) Answer : (2)

Solution:

Reticulate venation is a characteristic of dicot leaf. In monocot leaf, parallel venation is found. Abaxial surface generally bears more stomata in dicot leaf and conjoint closed vascular bundles are seen.

(92) Answer : (3)

Solution:

X → Monocot stem, Y → Roots of dicot or monocot plants

In monocot stem, hypodermis is sclerenchymatous, endodermis is absent and vascular bundles are of endarch type. In roots, exarch type of vascular bundles are found and hypodermis is absent

(93) Answer : (4)

Solution:

Stacking of one base pair over the other in double helix, hydrogen bonding and presence of thymine instead of uracil, all confer additional stability to DNA

(94) Answer : (1)

Solution:

Nucleases are enzymes that cleave the bond between nucleotides in nucleic acids. RNA polymerase II transcribes hnRNA. Severo Ochoa enzyme polymerises RNA.

(95) Answer : (4)

Solution:

DNA synthesis always occurs in 5' to 3' direction for both leading and lagging strands. RNA primers are required to initiate DNA synthesis because DNA polymerase cannot start the synthesis on its own.

(96) Answer : (3)

Solution:

AUG codes for the amino acid methionine.

(97) Answer : (2)

Solution:

The given structure is of vessel. Vessel is a long cylindrical tube-like structure made up of many cells called vessel members. Vessel members are interconnected through perforations in their common walls.

(98) Answer : (3)

Solution:

Avery, MacLeod and McCarty with the help of their experiments concluded that DNA is the genetic material.

(99) Answer : (1)

Solution:

Family has a group of related genera with still less number of similarities as compared to genus and species.

(100) Answer : (1)

Solution:

Mycelium of imperfect fungi is septate and branched. Most of them are decomposers of litter and help in mineral cycling.

(101) Answer : (2)

Solution:

Methanogens are the prokaryotes responsible for the production of biogas from the dung of ruminant animals.

(102) Answer : (1)

Solution:

Mustard has hypogynous flowers. While plum, peach and rose have perigynous flowers.

(103) Answer : (1)

Solution:

Plant family	Examples
• Compositae	– Marigold
• Solanaceae	– Chilli
• Fabaceae	– Sunhemp

• Malvaceae — Cotton

(104) Answer : (3)

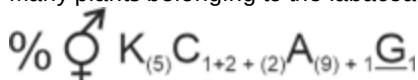
Solution:

Aristotle classified animals into Anaima and Enaima.

(105) Answer : (1)

Solution:

Many plants belonging to the fabaceae family are used as source of pulses. Floral formula of fabaceae family is:



(106) Answer : (4)

Solution:

Visceral nervous system is the part of the peripheral nervous system that comprises the whole complex of nerves, fibres, ganglia, and plexuses by which impulses travel from the central nervous system to the viscera and from the viscera to the central nervous system.

(107) Answer : (4)

Solution:

In gel electrophoresis, the DNA fragments separate (resolve) according to their size through the sieving effect provided by the agarose gel.

The positive electrode is called the anode. The negatively charged DNA moves towards it, thus DNA molecule A, being the smallest in size, will be found closer to the anode.

(108) Answer : (4)

Solution:

Exonucleases can cut the chromosomal DNA, but they will not act on the plasmids as plasmids are circular with no free ends.

(109) Answer : (4)

Solution:



rop codes for the proteins involved in the replication of the plasmid.

(110) Answer : (3)

Solution:

Knee joint is a hinge joint. Two halves of the pelvic girdle meet ventrally to form the pubic symphysis containing fibrous cartilage.

Pivot joint is present in between atlas and axis vertebrae and fibrous joint is present in between flat skull bones.

(111) Answer : (2)

Solution:

The fluid present in lymphatic system is called lymph. Tissue fluid has same mineral distribution as that in plasma.

Exchange of nutrients, gases, etc., between the blood and the cells always occur through this fluid. Fats are absorbed through lymph in the lacteals present in the intestinal villi.

(112) Answer : (2)

Solution:

Pituitary gland is located in a bony cavity called sella tursica and is attached to hypothalamus. Below the acromion is a depression called the glenoid cavity.

(113) Answer : (4)

Solution:

Ribs, scapula, sternum and skull bones are flat bones.

(114) Answer : (1)

Solution:

Plasma contains small amount of minerals like Na^+ , Ca^{++} , Mg^{++} , glucose, amino acids, etc., Plasma without the clotting factors is called serum.

(115) Answer : (3)**Solution:**

Hormones which interact with intracellular receptors (e.g. steroid hormones, iodothyronine, etc.) mostly regulate gene expression or chromosome function by the interaction of hormone-receptor complex.

Cortisol, testosterone, estradiol and progesterone are steroid hormones.

(116) Answer : (2)**Solution:**

The gymnosperms are the plants in which ovules are not enclosed within the ovary wall. The pollen grain germinates and pollen tube release the male gametes into the ovule, where it fuses with the egg cell in archegonia.

(117) Answer : (4)**Solution:**

The characteristic red colour in rhodophyceae members is due to the predominance of the red pigment, r-phycoerythrin.

(118) Answer : (2)**Solution:**

Acetobacter aceti is commercially used to synthesise acetic acid that is used in preparation of vinegar.

(119) Answer : (4)**Solution:**

The correct representation for population interactions are

	Species A	Species B
Competition :	–	–
Amensalism	–	0
Parasitism	–	+

(120) Answer : (4)**Solution:**

Bryophytes are non-vascular terrestrial plants of moist habitats in which a multicellular diploid sporophyte lives as a parasite on an independent multicellular haploid gametophyte. The dominant phase of the main plant body is a free-living gametophyte. The sporophyte consists of three parts namely capsule, seta and foot.

(121) Answer : (3)**Solution:**

The nodules of the roots of leguminous plants are formed by the symbiotic association of *Rhizobium*. The bacteria fix atmospheric N_2 into organic forms.

(122) Answer : (3)**Solution:**

Natality refers to the number of births during a given period in the population that are added to the initial density.

(123) Answer : (2)**Solution:**

Alexander Von Humboldt observed that within a region species richness increased with increasing explored area, but only up to a limit. In fact, the relation between species richness and area, for a wide variety of taxa turns out to be a rectangular hyperbola. Also on log scale, the slope of the line becomes steeper for larger areas.

(124) Answer : (4)**Solution:**

Oxygen, pollination and erosion control are broadly utilitarian while firewood comes under narrowly utilitarian ecosystem services of biodiversity.

(125) Answer : (1)**Solution:**

Mendel proposed that the 'factors' regulate the characters. These factors which are now known as genes are found in pairs called alleles.

(126) Answer : (2)**Solution:**

Linkage refers to the physical association of genes on a chromosome.

(127) Answer : (1)

Solution:

In incomplete dominance, F_1 progenies did not resemble either of the parents and was in between the two.

F_2 phenotypic and genotypic ratio in case of incomplete dominance (shown by starch grain size in pea plant) is 1 : 2 : 1.

(128) Answer : (3)

Solution:

Ovary is the basal bulged part of the pistil. Endothecium is anther wall layer below the epidermis.

(129) Answer : (2)

Solution:

Synergid is present in the embryo sac and is responsible for assisting the entry of pollen tube into the ovule.

(130) Answer : (3)

Solution:

Xenogamy brings genetic variability in the plants as it involves the transfer of pollens from one plant to the stigma of the other plant of the same species.

(131) Answer : (1)

Solution:

For treatment, cancer patients are given substances called biological response modifiers such as α -interferons which activate their immune system and help in destroying the tumor.

(132) Answer : (2)

Solution:

Acid in stomach, saliva in mouth, tears from eyes are physiological barriers of innate immunity.

(133) Answer : (1)

Solution:

Cerebellum, a part of hind brain integrates information received from semicircular canals of the ear and the auditory system.

Medulla oblongata controls cardiovascular reflexes.

Hypothalamus along with limbic system is involved in the regulation of sexual behaviour.

(134) Answer : (2)

Solution:

Glucagon	Insulin
• Secreted by α -cells	• Secreted by β -cells
• Peptide hormone	• Peptide hormone
• Stimulates glycogenolysis	• Stimulates glycogenesis
• Is hyperglycemic	• Is hypoglycemic
• Reduces cellular glucose uptake and utilisation	• Enhances cellular glucose uptake and utilisation.

(135) Answer : (1)

Solution:

If columnar or cuboidal cells bear cilia on their free surface, they are called ciliated epithelium. Their function is to move particles or mucus in a specific direction over the epithelium. They are mainly present in the inner surface of hollow organs like bronchioles and fallopian tubes.

(136) Answer : (2)

Solution:

During post-industrialisation period, the tree trunk became dark due to industrial smoke and soots. Under this condition the white-winged moths did not survive due to predation, dark-winged moths survived. So, natural selection is shifted in one direction. Hence, it shows directional selection.

(137) Answer : (4)

Solution:

Pre-historic cave art developed about 18,000 years ago. One such cave paintings by pre-historic humans can be seen at Bhimbetka rock shelter in Raisen district of Madhya Pradesh.

(138) Answer : (3)

Solution:

The epithelial cells of Bowman's capsule called podocytes are arranged in an intricate manner so as to leave some minute spaces called filtration slits or slit pores.

(139) Answer : (1)

Solution:

Bufo, *Naja* and *Vipera* all have tympanum and three chambered heart (2 auricle and 1 ventricle), but except *Bufo*, they all are limbless.

(140) Answer : (4)

Solution:

When we see objects in our immediate surroundings, we see them instantly and hence in the present time. Huge clusters of galaxies comprise the universe which is almost 20 billion years old.

(141) Answer : (4)

Solution:

Thorns of *Bougainvillea* and tendrils of *Cucurbita* represent homology which is based on divergent evolution. Wings of butterfly and that of birds represent analogy which are based on convergent evolution.

(142) Answer : (3)

Solution:

Glycosuria – Presence of glucose in urine.
Gout – Accumulation of uric acid crystals in joints.
Renal calculi – Mass of crystallised salts within the kidney.
Glomerulonephritis – Inflammation of glomeruli.

(143) Answer : (2)

Solution:

Adenosine deaminase (ADA) deficiency causes SCID (Severe Combined Immune Deficiency Disorder). This enzyme (ADA) is crucial for the immune system to function.

(144) Answer : (4)

Solution:

The Indian Government has set up organisations such as GEAC (Genetic Engineering Approval Committee) which will make decisions regarding the validity of GM research and the safety of introducing GM-organisms for public services.

(145) Answer : (2)

Solution:

Aurelia or jelly fish is a coelenterate. The characteristic feature of coelenterates is presence of cnidocytes. Choanocytes or collar cells are present in porifers. Comb plates are seen in ctenophores.

(146) Answer : (3)

Solution:

Basmati rice is distinct for its unique aroma and flavour and 27 documented varieties of Basmati are grown in India.

(147) Answer : (1)

Solution:

- *Aplysia* (mollusc) has organ system level of organisation.
- *Meandrina* (Coelenterate) has incomplete alimentary canal (blind sac body plan).

(148) Answer : (2)

Solution:

Blood is a fluid connective tissue containing plasma, red blood cells (RBCs), white blood cells (WBC) and platelets. It is the main circulating fluid that helps in the transport of various substances.

(149) Answer : (4)

Solution:

The activity of an enzyme can be affected by a change in the conditions which can alter the tertiary structure of the protein. These include temperature, pH, change in substrate concentration or binding of specific chemicals that regulate its activity.

(150) Answer : (3)

Solution:

Adenosine, guanosine, thymidine, uridine and cytidine are nucleosides. Uridine is present in RNA while thymidine is present in DNA. The genetic material of HIV is ssRNA. Hence, thymidine is absent.

(151) Answer : (3)

Solution:

The given figure represents β -plated sheet of secondary structure of protein. A protein thread does not exist throughout as an extended rigid rod. The thread is folded in the form of a helix (similar to a revolving staircase). Of course, only some portions of the protein thread are arranged in the form of a helix. In proteins, only right-handed helices are observed. Other regions of the protein thread are folded into other forms in what is called the secondary structure.

(152) Answer : (4)

Solution:

Chemical compounds undergo two types of changes. A physical change simply refers to a change in shape without breaking of bonds. This is a physical process. However, when bonds are broken and new bonds are formed during transformation, this will be called a chemical reaction.

(153) Answer : (3)

Solution:

Skeletal muscle tissue is closely attached to skeletal bones. In a typical muscle such as the biceps, striated (striped) skeletal muscle fibres are bundled together in a parallel fashion. Cardiac muscle tissue is a contractile tissue present only in the heart. Cell junctions fuse the plasma membranes of cardiac muscle cells and make them stick together.

(154) Answer : (3)

Solution:

Ovarian follicles in different stages of development are found embedded in ovarian stroma. The outermost covering of ovary is a thin epithelium.

(155) Answer : (2)

Solution:

Parturition is induced by a complex neuro-endocrine mechanism involving cortisol, estrogens and oxytocin.

(156) Answer : (2)

Solution:

Gibberellins play role in seed germination as well as in stem elongation.

(157) Answer : (4)

Solution:

Ethylene promotes leaf abscission in plants.

(158) Answer : (4)

Solution:

The principle of independent assortment applies to the genes located on different chromosomes and genes located on the same chromosome tend to be inherited together unless crossing over occurs.

(159) Answer : (3)

Solution:

Producers are also known as converters or transducers.

Primary consumers are key industry animals that convert plant matter into animal matter.

Primary carnivores are secondary consumers e.g. frogs.

Tertiary consumers are third order consumers e.g. snakes.

(160) Answer : (4)

Solution:

The first irreversible reaction of glycolysis where glucose is converted to glucose-6-phosphate is catalysed by hexokinase.

(161) Answer : (2)

Solution:

For fixation of each molecule of CO_2 into glucose, C_4 plants require additional 2ATP in comparison to C_3 plants.

For the formation of one sucrose molecule, 12 CO_2 molecules are required.

(162) Answer : (1)

Solution:

In C_4 plants, protorespiration does not occur. This is because those plants have a mechanism that increases the concentration of CO_2 at the enzyme site.

(163) Answer : (2)

Solution:

Based on the source of its nutrition or food, an organism occupies a specific place in the food chain that is known as its trophic level.

(164) Answer : (3)

Solution:

When fats are used as respiratory substrate, the RQ is less than 1 because amount of oxygen consumed is more than the amount of CO_2 evolved.

(165) Answer : (1)

Solution:

Na^+/K^+ pump is an example of active transport.

(166) Answer : (2)

Solution:

- (i) Microfilaments help in pseudopodia formation.
- (ii) Enzymes and electron carriers for the formation of ATP are present in the inner membrane of mitochondria.

(167) Answer : (3)

Solution:

Centriole duplication occurs during the S phase of cell cycle.

(168) Answer : (2)

Solution:

Tetrad formation is a characteristic feature of zygotene stage.

(169) Answer : (2)

Solution:

(i) Quiescent stage – G_0 stage

(ii) Metaphase is the best stage to study the morphology of chromosomes.

(iii) Diplotene is the stage, where dissolution of synaptonemal complex takes place.

(iv) Crossing over occurs in pachytene stage.

(170) Answer : (4)

Solution:

Nucleoplasm contains highly extended and elaborate nucleoprotein fibres called chromatin.

(171) Answer : (4)

Solution:

Frogs are beneficial for mankind because they eat insects and protect crops. Frogs serve as an important link of food chain and food web in ecosystem.

(172) Answer : (3)

Solution:

The placenta is connected to embryo through an umbilical cord which helps in transport of substances to and from the embryo.

(173) Answer : (3)

Solution:

During fertilisation, sperm attaches with its receptor present on zona pellucida layer of ovum to induce changes in the membrane that blocks the entry of additional sperms and thus ensures entry of only one sperm in ovum to maintain diploid number of chromosomes.

(174) Answer : (2)

Solution:

The abdomen in both males and females consist of 10 segments. In females, the 7th sternum is boat shaped and together with 8th and 9th sterna forms a brood or genital pouch.

(175) Answer : (1)

Solution:

Utilisation of O_2 by the cells for catabolic reactions and resultant release of CO_2 is the result of cellular respiration. The solubility of CO_2 is 20-25 times higher than that of O_2 . The volume of air involved in breathing movements can be estimated by using a spirometer which helps in clinical assessment of pulmonary functions.

(176) Answer : (2)

Solution:

Relaxation of the diaphragm and the inter-costal muscles returns the diaphragm and sternum to their normal positions and reduce the thoracic volume and thereby the pulmonary volume. This leads to an increase in intra-pulmonary pressure to slightly above the atmospheric pressure causing the expulsion of air from the lungs, i.e., expiration.

(177) Answer : (4)

Solution:

India was amongst the first countries in the world to initiate action plans and programmes at a national level to attain total reproductive health as a social goal.

(178) Answer : (3)

Solution:

STIs could lead to complications in later stages which include pelvic inflammatory diseases (PID), abortions, still births, ectopic pregnancies, infertility or even cancer of the reproductive tract.

(179) Answer : (1)

Solution:

Intra cytoplasmic sperm injection (ICSI) is a specialised procedure to form an embryo in the laboratory in which a sperm is directly injected into the ovum. Infertility cases either due to inability of the male partner to inseminate the female or due to

very low sperm counts in the ejaculate, could be corrected by artificial insemination (AI) technique.

(180) Answer : (2)

Solution:

Lower invertebrates like sponges, coelenterates, flatworms, *etc.*, exchange gases by simple diffusion over their entire body surface. Vascularised bags called lungs (pulmonary respiration) are used by the terrestrial animals for the exchange of gases. Among vertebrates, fishes use gills whereas amphibians, reptiles, birds and mammals respire through lungs.

